IE 622
Detailed Solutions — Practice Problem Set 12

Key Concepts Used in This Problem Set

This section summarises the central definitions and theorems that appear throughout the problems. Reading this first will make every proof below easier to follow.

1. Conditional Expectation

Definition. Let $(\Omega, \mathcal{F}, P)$ be a probability space, X an integrable r.v., and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. The *conditional expectation* $E[X \mid \mathcal{G}]$ is the (a.s. unique) random variable Z such that:

1. Z is $\mathcal{G}$-measurable.
2. $\displaystyle\int_G Z\, dP = \int_G X\, dP$ for every $G \in \mathcal{G}$ (the *partial-averaging* property).

Key properties used in the solutions.

- **Tower property**: If $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{F}$, then $E\big[E[X \mid \mathcal{B}] \mid \mathcal{A}\big] = E[X \mid \mathcal{A}]$ a.s.
- **Pulling out knowns**: If Y is $\mathcal{G}$-measurable and $E|XY| < \infty$, then $E[XY \mid \mathcal{G}] = Y \cdot E[X \mid \mathcal{G}]$ a.s.
- **Measurability**: If X is $\mathcal{G}$-measurable, then $E[X \mid \mathcal{G}] = X$ a.s.
- **Independence**: If $X \perp \mathcal{G}$, then $E[X \mid \mathcal{G}] = E[X]$ a.s.

2. Filtrations

A *filtration* is an increasing sequence of σ-algebras $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}$. Intuitively, $\mathcal{F}_n$ represents the information available up to time n. The *natural filtration* of a process $\{X_n\}$ is $\mathcal{F}_n^X = \sigma(X_1, \ldots, X_n)$.

3. Martingales, Supermartingales, and Submartingales

Definition. An adapted, integrable process $\{M_n, \mathcal{F}_n\}$ is a:

- **Martingale** if $E[M_{n+1} \mid \mathcal{F}_n] = M_n$ a.s. for all n.
- **Supermartingale** if $E[M_{n+1} \mid \mathcal{F}_n] \leq M_n$ a.s. for all n.
- **Submartingale** if $E[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ a.s. for all n.

Intuition. A martingale models a *fair game*: your expected future wealth equals your current wealth. A supermartingale models an *unfair game* (losing on average), and a submartingale a winning one. The classic example: M_n = fortune after n steps of a fair coin-flip game is a martingale.

Standard martingales used in this set.

1. $\{X_n\}$ = symmetric random walk ($E[X_{n+1} \mid \mathcal{F}_n] = X_n$).
2. $\{X_n^2 - n\}$ = squared symmetric random walk minus time (since $\mathrm{Var}(Z_i) = 1$).
3. $\{\mu^{-n} X_n\}$ in Galton–Watson process.
4. $\{R_n/(n + r + b)\}$ in Pólya's urn.

4. The Optional Stopping Theorem (OST)

Statement. Let $\{M_n, \mathcal{F}_n\}$ be a martingale and T a stopping time. Under any of the following sufficient conditions:

(a) T is bounded ($T \leq N$ for some fixed N), or

(b) The stopped process $\{M_{n\wedge T}\}$ is dominated by an integrable r.v. and $T < \infty$ a.s., or

(c) $E[T] < \infty$ and increments $|M_{n+1} - M_n|$ are bounded,

we have $E[M_T] = E[M_0]$.

Why it's powerful. OST lets us evaluate expectations at random (stopping) times by relating them to the initial value. In Problems 8(a) and 8(b), this is the core tool.

5. Markov Property of DTMCs

A DTMC $\{X_n\}$ satisfies the *Markov property*:

$$P(X_{n+1} = y \mid X_0, \ldots, X_n) = P(X_{n+1} = y \mid X_n) = p_{X_n,y}.$$

Equivalently, for any measurable g,

$$E[g(X_{n+1}) \mid \mathcal{F}_n] = E[g(X_{n+1}) \mid X_n] = \sum_y p_{X_n,y}\, g(y).$$

This is used in Problems 3, 5, and 6 to simplify conditional expectations.

6. Generator of a Markov Chain

For a DTMC with transition matrix P, the *(discrete) generator* (or Dynkin operator) acting on $f : \mathcal{S} \to \mathbb{R}$ is

$$(\mathcal{L}f)(x) = \sum_y p_{x,y}(f(y) - f(x)) = E[f(X_1) - f(X_0) \mid X_0 = x].$$

This measures the *expected drift* of f in one step. When $(\mathcal{L}f)(x) = 0$ for all x, f is called *harmonic*, and $\{f(X_n)\}$ is a martingale.

7. Superharmonic Functions and the Mean Value Property

A function $f : \mathbb{R}^d \to \mathbb{R}$ is *superharmonic* if it satisfies the sub-mean-value property:

$$f(x) \geq \frac{1}{\mathrm{vol}(B(x,r))} \int_{B(x,r)} f(y)\, dy \quad \forall\, x, r > 0.$$

Intuitively: the value of f at any point is at least the average of f over any surrounding ball. Harmonic functions (like $f(x) = x$ in 1D, or $f(x) = \log|x|$ in 2D for $x \neq 0$) satisfy equality. The connection to supermartingales is explored in Problem 5.

8. Doob Decomposition (background)

Any adapted integrable process $\{X_n\}$ can be uniquely written as $X_n = M_n + A_n$ where $\{M_n\}$ is a martingale and $\{A_n\}$ is a predictable process (i.e., A_n is $\mathcal{F}_{n-1}$-measurable) with $A_0 = 0$. In Problem 3, the sum $\sum_{i=0}^{n-1} (\mathcal{L}f)(X_i)$ serves as the predictable compensator A_n, and $Y_n = f(X_n) - f(X_0) - A_n$ is the martingale part.

Problem 1

Problem statement. Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose $\{B_i\}_{i\geq 1} \subset \mathcal{F}$ is a disjoint partition of Ω (i.e., $\Omega = \bigcup_{i=1}^{\infty} B_i$ and $B_i \cap B_j = \emptyset$ for $i \neq j$) with $P(B_i) > 0$ for all i. Let $\mathcal{G} = \sigma(B_1, B_2, \ldots)$. For an $\mathcal{F}$-measurable, integrable random variable X, show that

$$E[X \mid \mathcal{G}] = \frac{E[X\mathbf{1}_{B_i}]}{P(B_i)} \quad \text{on } B_i.$$

Background: definition of conditional expectation. Recall that $Z = E[X \mid \mathcal{G}]$ is the *unique* (a.s.) $\mathcal{G}$-measurable, integrable random variable satisfying

$$\int_G Z\, dP = \int_G X\, dP \quad \text{for every } G \in \mathcal{G}.$$

We will verify that $Z := \sum_{i=1}^{\infty} \frac{E[X\mathbf{1}_{B_i}]}{P(B_i)} \mathbf{1}_{B_i}$ satisfies both conditions.

Solution. Step 1 — Describing the σ-algebra $\mathcal{G}$. Because $\{B_i\}$ is a partition of Ω, the atoms of $\mathcal{G} = \sigma(B_1, B_2, \ldots)$ are exactly the sets B_i themselves. Any $G \in \mathcal{G}$ is a (possibly empty, possibly infinite) union $G = \bigcup_{j\in J} B_j$ for some $J \subseteq \mathbb{N}$. This follows because: (a) every such union is a member of $\mathcal{G}$; (b) the class of such unions is closed under complements ($\Omega \setminus \bigcup_{j\in J} B_j = \bigcup_{j\notin J} B_j$) and countable unions, hence is itself a σ-algebra containing every B_i, making it equal to $\mathcal{G}$.

Step 2 — Defining the candidate Z. For each $\omega \in \Omega$ there is a unique index $i(\omega)$ with $\omega \in B_{i(\omega)}$. Define

$$Z(\omega) := \frac{E[X\mathbf{1}_{B_{i(\omega)}}]}{P(B_{i(\omega)})},$$

or equivalently,

$$Z = \sum_{i=1}^{\infty} c_i\, \mathbf{1}_{B_i}, \quad c_i := \frac{E[X\mathbf{1}_{B_i}]}{P(B_i)}.$$

Note: $E[X\mathbf{1}_{B_i}]$ is a real number (it is a finite constant because X is integrable and $\mathbf{1}_{B_i} \le 1$, so $|E[X\mathbf{1}_{B_i}]| \le E[|X|\mathbf{1}_{B_i}] < \infty$). Integrability of Z:

$$E|Z| = \sum_{i=1}^{\infty} |c_i|\, P(B_i) = \sum_{i=1}^{\infty} |E[X\mathbf{1}_{B_i}]| \le \sum_{i=1}^{\infty} E[|X|\mathbf{1}_{B_i}] = E|X| < \infty.$$

Step 3 — Measurability. Each indicator $\mathbf{1}_{B_i}$ is $\mathcal{G}$-measurable (since $B_i \in \mathcal{G}$). A countable linear combination of $\mathcal{G}$-measurable functions (with deterministic coefficients c_i) is $\mathcal{G}$-measurable. Hence Z is $\mathcal{G}$-measurable.

Step 4 — Partial averaging (the key verification). Take any $G \in \mathcal{G}$. By Step 1, $G = \bigcup_{j\in J} B_j$ for some $J \subseteq \mathbb{N}$. The sets $\{B_j\}_{j\in J}$ are disjoint, so

$$\int_G Z\, dP = E[Z\,\mathbf{1}_G] = E\left[\left(\sum_{i=1}^{\infty} c_i\,\mathbf{1}_{B_i}\right)\mathbf{1}_{\bigcup_{j\in J} B_j}\right].$$

Since $\mathbf{1}_{B_i} \cdot \mathbf{1}_{\bigcup_{j\in J} B_j} = \mathbf{1}_{B_i}$ if $i \in J$ and 0 otherwise, the sum collapses to $j \in J$ only:

$$= E\left[\sum_{j\in J} c_j\,\mathbf{1}_{B_j}\right] = \sum_{j\in J} c_j\, E[\mathbf{1}_{B_j}] \quad \text{(Fubini/MCT, since terms are non-negative after splitting } X = X^+ - X^-\text{)}$$

$$= \sum_{j\in J} \frac{E[X\mathbf{1}_{B_j}]}{P(B_j)} \cdot P(B_j) = \sum_{j\in J} E[X\mathbf{1}_{B_j}] = E\left[X \sum_{j\in J} \mathbf{1}_{B_j}\right] = E[X\,\mathbf{1}_G] = \int_G X\, dP.$$

(The interchange of expectation and sum is justified because $\sum_{j\in J} E[|X|\mathbf{1}_{B_j}] \le E|X| < \infty$.)

Conclusion. Z is $\mathcal{G}$-measurable, integrable, and satisfies the partial-averaging property for every $G \in \mathcal{G}$. By uniqueness of conditional expectation (a.s.),

$$E[X \mid \mathcal{G}] = Z = \frac{E[X\mathbf{1}_{B_i}]}{P(B_i)} \quad \text{on } B_i. \qquad \square$$

Problem 2

Problem statement. Suppose $\{X_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$. Let $\mathcal{G}_n = \sigma(X_1, \ldots, X_n)$. Show that $\{X_n\}$ is a martingale with respect to $\{\mathcal{G}_n\}$.

Key observations.

1. $\mathcal{G}_n \subseteq \mathcal{F}_n$: each X_k for $k \le n$ is $\mathcal{F}_k \subseteq \mathcal{F}_n$-measurable, so $\sigma(X_1, \ldots, X_n) \subseteq \mathcal{F}_n$.
2. The filtration $\{\mathcal{G}_n\}$ is increasing: $\mathcal{G}_n \subseteq \mathcal{G}_{n+1}$ since $\sigma(X_1, \ldots, X_n) \subseteq \sigma(X_1, \ldots, X_n, X_{n+1})$.

Solution. (i) Integrability. Since $\{X_n\}$ is a martingale w.r.t. $\{\mathcal{F}_n\}$, $E|X_n| < \infty$ for all n. So X_n is integrable.

(ii) Adaptedness. By definition of $\mathcal{G}_n = \sigma(X_1, \ldots, X_n)$, the random variable X_n is $\mathcal{G}_n$-measurable (it is one of the generating variables).

(iii) Martingale property. We must show $E[X_{n+1} \mid \mathcal{G}_n] = X_n$ a.s.

Step 1: Apply the tower property. Since $\mathcal{G}_n \subseteq \mathcal{F}_n$,

$$E[X_{n+1} \mid \mathcal{G}_n] = E\big[E[X_{n+1} \mid \mathcal{F}_n] \mid \mathcal{G}_n\big].$$

(The tower property states: if $\mathcal{A} \subseteq \mathcal{B}$ are sub-σ-algebras, then $E[E[Y \mid \mathcal{B}] \mid \mathcal{A}] = E[Y \mid \mathcal{A}]$.)

Step 2: Since $\{X_n\}$ is a martingale w.r.t. $\{\mathcal{F}_n\}$,

$$E[X_{n+1} \mid \mathcal{F}_n] = X_n \quad \text{a.s.}$$

Step 3: Substituting,

$$E[X_{n+1} \mid \mathcal{G}_n] = E[X_n \mid \mathcal{G}_n].$$

Step 4: X_n is $\mathcal{G}_n$-measurable (Step (ii)), so $E[X_n \mid \mathcal{G}_n] = X_n$ a.s. (a measurable random variable is its own conditional expectation given its own σ-algebra).

Combining:

$$E[X_{n+1} \mid \mathcal{G}_n] = X_n \quad \text{a.s.}$$

Therefore $\{X_n\}$ is a martingale with respect to $\{\mathcal{G}_n\}$. $\square$

Remark. The key structural fact is that $\mathcal{G}_n \subseteq \mathcal{F}_n$: passing to a *coarser* filtration preserves the martingale property. The $\mathcal{G}_n$-martingale property then follows by smoothing over the extra information in $\mathcal{F}_n$.

Problem 3 (Dynkin's Martingale)

Problem statement. Let $\{X_n\}$ be a DTMC on state space $\mathcal{S}$ with transition matrix P. Define the *generator* operator

$$(\mathcal{L}f)(x) := \sum_{y \in \mathcal{S}} p_{x,y}(f(y) - f(x)) = E[f(X_1) - f(X_0) \mid X_0 = x].$$

Set $Y_0 = 0$ and, for $n \geq 1$,

$$Y_n := f(X_n) - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i).$$

Show that $\{Y_n\}$ is a martingale with respect to $\mathcal{H}_n := \sigma(X_0, \ldots, X_n)$.

Intuition. The sum $\sum_{i=0}^{n-1} (\mathcal{L}f)(X_i)$ is the *predictable compensator* of the process $\{f(X_n)\}$, i.e., it removes the expected drift so that the residual is a martingale. This construction mirrors the Doob decomposition.

Solution. **(i) Adaptedness.** Y_n depends only on $X_0, X_1, \ldots, X_n$, hence is $\mathcal{H}_n$-measurable.

(ii) Integrability. Assume f is bounded (or that $E|f(X_n)| < \infty$ for all n; since $(\mathcal{L}f)(x) = \sum_y p_{x,y} f(y) - f(x)$ is a finite linear combination of values of f, finiteness of $E|f(X_k)|$ implies finiteness of $E|(\mathcal{L}f)(X_k)|$). Then $E|Y_n| < \infty$.

(iii) Martingale property. Compute the increment:

$$Y_{n+1} - Y_n = \Big[f(X_{n+1}) - f(X_0) - \sum_{i=0}^{n} (\mathcal{L}f)(X_i)\Big] - \Big[f(X_n) - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i)\Big] = f(X_{n+1}) - f(X_n) - (\mathcal{L}f)(X_n).$$

Now take the conditional expectation given $\mathcal{H}_n$:

$$E[Y_{n+1} - Y_n \mid \mathcal{H}_n] = E[f(X_{n+1}) \mid \mathcal{H}_n] - f(X_n) - (\mathcal{L}f)(X_n).$$

By the Markov property of $\{X_n\}$,

$$E[f(X_{n+1}) \mid \mathcal{H}_n] = E[f(X_{n+1}) \mid X_n] = \sum_{y \in \mathcal{S}} p_{X_n,y}\, f(y).$$

Note that

$$(\mathcal{L}f)(X_n) = \sum_{y \in \mathcal{S}} p_{X_n,y}(f(y) - f(X_n)) = \sum_{y \in \mathcal{S}} p_{X_n,y} f(y) - f(X_n) \sum_{y \in \mathcal{S}} p_{X_n,y} = \sum_{y \in \mathcal{S}} p_{X_n,y} f(y) - f(X_n),$$

since $\sum_y p_{X_n,y} = 1$. Therefore,

$$E[f(X_{n+1}) \mid \mathcal{H}_n] = (\mathcal{L}f)(X_n) + f(X_n).$$

Substituting back:

$$E[Y_{n+1} - Y_n \mid \mathcal{H}_n] = \big[(\mathcal{L}f)(X_n) + f(X_n)\big] - f(X_n) - (\mathcal{L}f)(X_n) = 0.$$

Therefore $E[Y_{n+1} \mid \mathcal{H}_n] = Y_n$ a.s., proving $\{Y_n\}$ is a martingale. $\square$

Problem 4 (Galton–Watson Branching Process)

Problem statement. Let $\{Z_i^n\}$ be i.i.d. nonneg. integer-valued r.v.'s with mean $\mu > 0$. Define $X_0 = 1$ and $X_{n+1} = \sum_{i=1}^{X_n} Z_i^{n+1}$ for $n \geq 0$. Let $\mathcal{F}_n = \sigma(Z^1, Z^2, \ldots, Z^n)$, where $Z^k = (Z_1^k, Z_2^k, \ldots)$. Show that $\{M_n\} := \{\mu^{-n} X_n, n \geq 0\}$ is a martingale w.r.t. $\{\mathcal{F}_n\}$.

Setup and notation. X_n = population size at generation n. Z_i^{n+1} = number of offspring of individual i in generation n. The key independence: the offspring counts $\{Z_i^{n+1}\}_{i \geq 1}$ are independent of $\mathcal{F}_n$ (they are "new" randomness in generation $n+1$).

Solution. **(i) Adaptedness.** X_n is determined by the first n generations' offspring:

$$X_1 = Z_1^1 \quad \text{(from the single ancestor)},$$
$$X_2 = \sum_{i=1}^{X_1} Z_i^2, \quad \ldots$$

So $X_n \in \mathcal{F}_n$, and hence $M_n = \mu^{-n} X_n \in \mathcal{F}_n$.

(ii) Integrability. We show by induction that $E[X_n] = \mu^n$, which gives $E[M_n] = 1 < \infty$.

Base case: $E[X_0] = 1 = \mu^0$.

Inductive step: Assume $E[X_n] = \mu^n$. Then

$$E[X_{n+1}] = E\left[\sum_{i=1}^{X_n} Z_i^{n+1}\right] = E\left[E\left[\sum_{i=1}^{X_n} Z_i^{n+1} \,\middle|\, X_n\right]\right].$$

Given $X_n = k$, the sum $\sum_{i=1}^{k} Z_i^{n+1}$ consists of k i.i.d. terms each with mean μ, so $E[\sum_{i=1}^{k} Z_i^{n+1}] = k\mu$. Hence

$$E\left[\sum_{i=1}^{X_n} Z_i^{n+1} \,\middle|\, X_n\right] = X_n \mu,$$

and

$$E[X_{n+1}] = E[X_n\mu] = \mu E[X_n] = \mu \cdot \mu^n = \mu^{n+1}.$$

(iii) Martingale property. We compute $E[M_{n+1} \mid \mathcal{F}_n]$:

$$\begin{aligned} E[M_{n+1} \mid \mathcal{F}_n] &= \mu^{-(n+1)} E[X_{n+1} \mid \mathcal{F}_n] \\ &= \mu^{-(n+1)} E\left[\sum_{i=1}^{X_n} Z_i^{n+1} \,\middle|\, \mathcal{F}_n\right]. \end{aligned}$$

Since X_n is $\mathcal{F}_n$-measurable and $\{Z_i^{n+1}\}_{i\geq 1}$ are independent of $\mathcal{F}_n$, we can condition on $X_n = k$ and use:

$$E\left[\sum_{i=1}^{X_n} Z_i^{n+1} \,\middle|\, \mathcal{F}_n\right] = \sum_{k=0}^{\infty} \mathbf{1}_{\{X_n=k\}}\, E\left[\sum_{i=1}^{k} Z_i^{n+1}\right] = \sum_{k=0}^{\infty} \mathbf{1}_{\{X_n=k\}} \cdot k\mu = X_n\mu.$$

(This is a standard application of: if N is $\mathcal{F}_n$-measurable and $\{W_i\}$ are i.i.d. independent of $\mathcal{F}_n$, then $E[\sum_{i=1}^{N} W_i \mid \mathcal{F}_n] = N \cdot EW_1$.) Therefore,

$$E[M_{n+1} \mid \mathcal{F}_n] = \mu^{-(n+1)} \cdot X_n\mu = \mu^{-n} X_n = M_n.$$

Hence $\{M_n\}$ is a martingale. $\square$

Remark. The martingale $\{M_n\}$ converges a.s. (by the martingale convergence theorem, since $M_n \geq 0$). The limit $M_\infty = \lim_{n\to\infty} \mu^{-n} X_n$ is related to the extinction probability: if $\mu \leq 1$, then $M_\infty = 0$ a.s.

Problem 5 (Superharmonic Functions)

Problem statement. A function $f : \mathbb{R}^d \to \mathbb{R}$ is *superharmonic* if, for all $x \in \mathbb{R}^d$ and all $r > 0$,

$$f(x) \geq \frac{1}{\mathrm{vol}(B(0,r))} \int_{B(x,r)} f(y)\, dy.$$

Let $\{Z_n\}$ be i.i.d., uniformly distributed on $B(0,1)$. Set $S_0 = 0$ and $S_n = S_{n-1} + Z_n$. Define $X_n = f(S_n)$. Show $\{X_n\}$ is a supermartingale with respect to $\mathcal{F}_n = \sigma(Z_1, \ldots, Z_n) = \sigma(S_1, \ldots, S_n)$.

Solution. (i) Adaptedness. $X_n = f(S_n)$. Since $S_n = Z_1 + \cdots + Z_n$ is $\sigma(Z_1, \ldots, Z_n) = \mathcal{F}_n$-measurable, so is $X_n = f(S_n)$.

(ii) Integrability. We assume f is integrable enough (e.g., $E|f(S_n)| < \infty$ for all n). This holds, e.g., if f is bounded.

(iii) Supermartingale property. We compute $E[X_{n+1} \mid \mathcal{F}_n]$.

Since $S_{n+1} = S_n + Z_{n+1}$ and Z_{n+1} is independent of $\mathcal{F}_n$ (it is the "new" increment), and S_n is $\mathcal{F}_n$-measurable:

$$E[X_{n+1} \mid \mathcal{F}_n] = E[f(S_n + Z_{n+1}) \mid \mathcal{F}_n].$$

Condition on $\mathcal{F}_n$: since $Z_{n+1} \perp \mathcal{F}_n$ and S_n is $\mathcal{F}_n$-measurable, the conditional distribution of $S_n + Z_{n+1}$ given $\mathcal{F}_n$ is the distribution of $S_n + Z_{n+1}$ with S_n frozen at its current value. That is, for fixed $s = S_n(\omega)$,

$$E[f(S_n + Z_{n+1}) \mid \mathcal{F}_n](\omega) = E[f(s + Z_{n+1})] = \int_{B(0,1)} f(s+z)\, \frac{dz}{\mathrm{vol}(B(0,1))}.$$

Now make the substitution $y = s + z$ (i.e., $z = y - s$). As z ranges over $B(0,1) = \{z : |z| \le 1\}$, y ranges over $\{y : |y - s| \le 1\} = B(s,1)$. The Jacobian is 1. Hence:

$$\int_{B(0,1)} f(s+z)\,\frac{dz}{\mathrm{vol}(B(0,1))} = \frac{1}{\mathrm{vol}(B(s,1))}\int_{B(s,1)} f(y)\,dy$$

where we use that $\mathrm{vol}(B(s,1)) = \mathrm{vol}(B(0,1))$ (both are balls of radius 1 in $\mathbb{R}^d$, just centered differently; their volume is the same).

By the *superharmonic* property of f applied at the point $x = s = S_n$ with $r = 1$:

$$f(s) \ge \frac{1}{\mathrm{vol}(B(s,1))}\int_{B(s,1)} f(y)\,dy.$$

Therefore,

$$E[X_{n+1} \mid \mathcal{F}_n] = \frac{1}{\mathrm{vol}(B(S_n,1))}\int_{B(S_n,1)} f(y)\,dy \le f(S_n) = X_n.$$

This holds a.s. (since it holds for every fixed realization of S_n).

Hence $E[X_{n+1} \mid \mathcal{F}_n] \le X_n$ a.s., so $\{X_n\}$ is a supermartingale. $\square$

Remark. If f is harmonic ($\ge$ replaced by $=$), then $\{X_n\}$ becomes a martingale. Superharmonic functions correspond to supermartingales, and subharmonic to submartingales — this is the deep connection between potential theory and martingale theory.

Problem 6

Problem statement. Let $\{X_n\}$ be a DTMC on $\mathcal{X}$ with transition probabilities $\{p_{x,y}\}$. Suppose $f : \mathcal{X} \to \mathbb{R}$ is bounded and satisfies

$$\sum_{y\in\mathcal{X}} p_{x,y} f(y) \le \lambda f(x)$$

for some $\lambda > 0$ and all $x \in \mathcal{X}$. Show that $\{N_n\} := \{\lambda^{-n} f(X_n)\}$ is a supermartingale with respect to $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$.

Interpretation. The condition $\sum_y p_{x,y} f(y) \le \lambda f(x)$ says: the expected value of f after one step from x is at most $\lambda f(x)$. This is the *Foster-Lyapunov* type condition. When we discount by λ^{-n}, the expected value after the discount is non-increasing.

Solution. (i) Adaptedness. $N_n = \lambda^{-n} f(X_n)$ is $\mathcal{F}_n$-measurable (it is a function of X_n, which is $\mathcal{F}_n$-measurable).

(ii) Integrability. Since f is bounded, say $|f| \le M$, we have $|N_n| \le \lambda^{-n} M < \infty$, so N_n is bounded and integrable.

(iii) Supermartingale property.

$$E[N_{n+1} \mid \mathcal{F}_n] = \lambda^{-(n+1)}\, E[f(X_{n+1}) \mid \mathcal{F}_n].$$

By the Markov property:

$$E[f(X_{n+1}) \mid \mathcal{F}_n] = E[f(X_{n+1}) \mid X_n] = \sum_{y\in\mathcal{X}} p_{X_n,y}\, f(y).$$

By the given condition applied at $x = X_n$:

$$\sum_{y \in \mathcal{X}} p_{X_n,y}\, f(y) \le \lambda\, f(X_n).$$

Therefore,

$$E[N_{n+1} \mid \mathcal{F}_n] \le \lambda^{-(n+1)} \cdot \lambda\, f(X_n) = \lambda^{-n} f(X_n) = N_n.$$

Hence $E[N_{n+1} \mid \mathcal{F}_n] \le N_n$ a.s., so $\{N_n\}$ is a supermartingale. $\square$

Remark. If $\lambda = 1$ and the inequality is an equality $\sum_y p_{x,y} f(y) = f(x)$, then f is a *harmonic function* for the chain and $\{f(X_n)\}$ is a martingale. If $\lambda = 1$ and $\le$ holds, f is superharmonic and $\{f(X_n)\}$ is a supermartingale. Problem 6 generalizes this to arbitrary $\lambda > 0$.

Problem 7 (Pólya's Urn)

Setup. Start with r red and b blue balls (total $r + b$, both $r, b > 0$). At each step: draw a ball uniformly at random, note its color, replace it, add a new ball of the same color. $R_n = \#$red balls after n draws.

Part (a): $Y_n = R_n/(n + r + b)$ is a martingale

Solution. **Setup.** After n steps, total balls $= n+r+b$, red balls $= R_n$, blue balls $= n+r+b-R_n$.

Adaptedness. $Y_n = R_n/(n + r + b)$ is $\mathcal{F}_n = \sigma(R_1, \ldots, R_n)$-measurable.

Integrability. $0 \le Y_n \le 1$, so $E|Y_n| < \infty$.

Martingale property. At step $n + 1$, given $\mathcal{F}_n$ (so R_n is known):

- With probability $\dfrac{R_n}{n + r + b}$, a red ball is drawn, so $R_{n+1} = R_n + 1$.
- With probability $\dfrac{(n + r + b) - R_n}{n + r + b}$, a blue ball is drawn, so $R_{n+1} = R_n$.

Therefore,

$$\begin{aligned} E[R_{n+1} \mid \mathcal{F}_n] &= (R_n + 1) \cdot \frac{R_n}{n + r + b} + R_n \cdot \frac{(n + r + b - R_n)}{n + r + b} \\ &= \frac{R_n^2 + R_n + R_n(n + r + b) - R_n^2}{n + r + b} \\ &= \frac{R_n(n + r + b) + R_n}{n + r + b} \\ &= \frac{R_n(n + r + b + 1)}{n + r + b}. \end{aligned}$$

Hence,

$$E[Y_{n+1} \mid \mathcal{F}_n] = \frac{E[R_{n+1} \mid \mathcal{F}_n]}{(n + 1) + r + b} = \frac{R_n(n + r + b + 1)}{(n + r + b)(n + r + b + 1)} = \frac{R_n}{n + r + b} = Y_n.$$

So $\{Y_n\}$ is a martingale. $\square$

Part (b): $E[(T+2)^{-1}] = 1/4$ when $r = b = 1$

Setup. With $r = b = 1$, the urn starts with 1 red and 1 blue ball (2 total). T = number of draws until the first blue ball.

Step 1: Distribution of T. At each draw, suppose the current number of red balls is k and the total number of balls is $k+1$ (since only red balls were added so far, having drawn all red for the first $k-1$ draws). The probability of drawing blue (the only blue ball) is $\frac{1}{k+1}$.

More carefully:

- At draw 1: urn has 1R, 1B. $P(\text{draw B}) = \frac{1}{2}$, so $P(T=1) = \frac{1}{2}$.
- If draw 1 is red: urn becomes 2R, 1B (total 3). $P(\text{draw B}) = \frac{1}{3}$.
- If draws 1,2 are red: urn becomes 3R, 1B (total 4). $P(\text{draw B at step 3}) = \frac{1}{4}$.

In general, $T = k$ means the first $k-1$ draws were all red and draw k is blue:

$$P(T=k) = \underbrace{\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{k-1}{k}}_{\text{prob. first } k-1 \text{ are red}} \cdot \underbrace{\frac{1}{k+1}}_{\text{draw } k \text{ is blue}}.$$

The numerator of the first product is $(k-1)!$ and the denominator is $2 \cdot 3 \cdots k = \frac{k!}{1!} = k!$, so the first product is $\frac{(k-1)!}{k!} = \frac{1}{k}$. Hence

$$\boxed{P(T=k) = \frac{1}{k(k+1)}}, \quad k = 1, 2, 3, \ldots$$

Verification: $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1$. ✓

Step 2: Computing $E[(T+2)^{-1}]$.

$$E\left[\frac{1}{T+2}\right] = \sum_{k=1}^{\infty} \frac{1}{k+2} \cdot P(T=k) = \sum_{k=1}^{\infty} \frac{1}{k+2} \cdot \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)}.$$

Step 3: Partial fractions for $\frac{1}{k(k+1)(k+2)}$. We decompose:

$$\frac{1}{k(k+1)(k+2)} = \frac{A}{k} + \frac{B}{k+1} + \frac{C}{k+2}.$$

Multiplying both sides by $k(k+1)(k+2)$:

$$1 = A(k+1)(k+2) + Bk(k+2) + Ck(k+1).$$

- $k = 0$: $1 = A \cdot 1 \cdot 2$, so $A = \frac{1}{2}$.
- $k = -1$: $1 = B(-1)(1)$, so $B = -1$.
- $k = -2$: $1 = C(-2)(-1)$, so $C = \frac{1}{2}$.

Therefore $\dfrac{1}{k(k+1)(k+2)} = \dfrac{1/2}{k} - \dfrac{1}{k+1} + \dfrac{1/2}{k+2}$.

Step 4: Summing the series.

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)} = \frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k} - \sum_{k=1}^{\infty}\frac{1}{k+1} + \frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k+2}.$$

Re-index: let $j = k+1$ in the second sum and $j = k+2$ in the third:

$$= \frac{1}{2}\sum_{k=1}^{\infty}\frac{1}{k} - \sum_{j=2}^{\infty}\frac{1}{j} + \frac{1}{2}\sum_{j=3}^{\infty}\frac{1}{j}.$$

Split the first sum as $\sum_{k=1}^{\infty}\frac{1}{k} = \frac{1}{1} + \frac{1}{2} + \sum_{j=3}^{\infty}\frac{1}{j}$, and the second sum as $\sum_{j=2}^{\infty}\frac{1}{j} = \frac{1}{2} + \sum_{j=3}^{\infty}\frac{1}{j}$. Let $S = \sum_{j=3}^{\infty}\frac{1}{j}$:

$$\begin{aligned}
&= \frac{1}{2}\left(1 + \frac{1}{2} + S\right) - \left(\frac{1}{2} + S\right) + \frac{1}{2}S \\
&= \frac{1}{2} + \frac{1}{4} + \frac{S}{2} - \frac{1}{2} - S + \frac{S}{2} \\
&= \frac{1}{4} + \left(\frac{S}{2} - S + \frac{S}{2}\right) = \frac{1}{4} + 0 = \frac{1}{4}.
\end{aligned}$$

Hence $E\left[(T+2)^{-1}\right] = \dfrac{1}{4}$. $\square$

Problem 8 (Symmetric Random Walk: Gambler's Ruin)

Setup. Z_i i.i.d., $P(Z_i = 1) = P(Z_i = -1) = \frac{1}{2}$. $X_0 = 0$, $X_n = \sum_{i=1}^{n} Z_i$. Fix $a, b \in \mathbb{Z}_{>0}$ and $T = \min\{n \geq 0 : X_n \in \{-a, b\}\}$.

Preliminary: $T < \infty$ a.s. and $E[T] < \infty$. By standard results on the symmetric random walk on $\mathbb{Z}$, the walk will hit any finite set of integers in finite expected time. In particular, $T < \infty$ a.s. and $E[T] < \infty$ (in fact $E[T] = ab$ as we will show).

Part (a): $P(X_T = -a) = b/(a+b)$

Martingale used. $\{X_n\}$ is itself a martingale, since $E[X_{n+1} \mid \mathcal{F}_n] = X_n + E[Z_{n+1}] = X_n + 0 = X_n$.

Verifying OST conditions. For $n \leq T$, we have $X_n \in (-a, b)$ (or $X_n \in \{-a, b\}$ when $n = T$), so $|X_{n\wedge T}| \leq \max(a, b)$. The stopped process $\{X_{n\wedge T}\}$ is bounded. By the Bounded Optional Stopping Theorem (or Doob's OST), since the stopped process is a bounded martingale and $T < \infty$ a.s.,

$$E[X_T] = E[X_0] = 0.$$

Computing the probability. Let $p = P(X_T = b)$ so $P(X_T = -a) = 1 - p$. Since $X_T \in \{-a, b\}$:

$$0 = E[X_T] = b \cdot p + (-a) \cdot (1-p) = bp - a + ap = (a+b)p - a.$$

Solving:

$$p = \frac{a}{a+b}, \qquad P(X_T = -a) = 1 - p = \frac{b}{a+b}. \qquad \square$$

Part (b): $E[T] = ab$

Key martingale. Consider $M_n = X_n^2 - n$. We verify it is a martingale:

$$\begin{aligned} E[M_{n+1} \mid \mathcal{F}_n] &= E[(X_n + Z_{n+1})^2 - (n+1) \mid \mathcal{F}_n] \\ &= E[X_n^2 + 2X_n Z_{n+1} + Z_{n+1}^2 \mid \mathcal{F}_n] - (n+1) \\ &= X_n^2 + 2X_n E[Z_{n+1}] + E[Z_{n+1}^2] - n - 1 \\ &= X_n^2 + 0 + 1 - n - 1 = X_n^2 - n = M_n. \end{aligned}$$

(We used $E[Z_{n+1}] = 0$, $E[Z_{n+1}^2] = 1$, and $Z_{n+1} \perp \mathcal{F}_n$.)

Applying OST. We apply OST to $\{M_n\}$ at time T. We need $E[M_T]$ to be well-defined. The stopped martingale $\{M_{n\wedge T}\}$ satisfies:

$$|M_{n\wedge T}| = |X_{n\wedge T}^2 - (n \wedge T)| \leq \max(a,b)^2 + n \wedge T \leq \max(a,b)^2 + T.$$

Since $E[T] < \infty$ (established in Part (a) context and confirmed below), by dominated convergence, $E[M_{n\wedge T}] \to E[M_T]$ as $n \to \infty$. Since $\{M_{n\wedge T}\}$ is a martingale, $E[M_{n\wedge T}] = E[M_0] = 0$ for all n. Hence $E[M_T] = 0$, i.e.,

$$E[X_T^2] = E[T].$$

Computing $E[X_T^2]$. We use $P(X_T = b) = a/(a+b)$ and $P(X_T = -a) = b/(a+b)$:

$$E[X_T^2] = b^2 \cdot \frac{a}{a+b} + a^2 \cdot \frac{b}{a+b} = \frac{ab^2 + a^2 b}{a+b} = \frac{ab(a+b)}{a+b} = ab.$$

Conclusion.

$$E[T] = E[X_T^2] = ab. \qquad \square$$

Remark on $E[T] < \infty$. To close the argument, we sketch why $E[T] < \infty$. Note $T \leq T'$ where $T' = \min\{n : X_n \notin (-a-b, a+b)\}$, and the random walk on a finite interval has $E[T'] < \infty$ by standard analysis (e.g., using the fact that at each step there is a positive probability of exiting a box of size $a+b$ within $a+b$ steps). From $E[T] = ab$ itself being finite (once derived), we also get the self-consistent conclusion.